

Requirements Document

Team 1

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Requirements Documents

Requirements Definition

FR_1: As a host, I want to be able to view the status of each table so I can keep track of which ones are available and which ones are not.

FR_2: As a host, I want to update the status of tables to indicate when they are occupied, empty, or need cleaning.

FR_3: As a host, I want to assign waiters to newly occupied tables so customers receive service quickly.

FR_4: As a waiter, I want to be able to view my assigned tables so I know which ones I am responsible for taking care of.

FR_5: As a waiter, I want to be able to view and update the status of tables to indicate when they are occupied, empty, or need cleaning.

FR_6: As a waiter, I want to be able to enter customer orders quickly to the kitchen staff so that they can be prepared immediately.

FR_7: As a waiter, I want to be able to view the status of customer orders so that I can give updates to customers on when to expect them.

FR_8: As a waiter, I want to be able to process payments so that customers can pay their tabs electronically or using cash.

FR_9: As a cook, I want to be able to see all upcoming orders in the queue so I know what to prepare next.

FR_10: As a cook, I want to be able to update the status of an order and give its ETA so that waiters know when to expect the order, so that they can notify the customers.

FR_11: As a cook, I want to be able to send a pop up alert to the assigned waiter of an order, letting them know that the order is ready so that it can be delivered promptly.

FR_12: As a busboy, I want to be able to view which tables are marked as 'dirty' so that I know which tables to clean and prepare.

FR_13: As a busboy, I want to be able to update the status of tables after they are cleaned so that the host knows that the table is available.

FR_14: As a manager, I want to be able to add, delete, and modify employee profiles so that I can manage restaurant staff efficiently.

FR_15: As a manager, I want to be able to track employee activity so that I can view employee work times, productivity, and performance.

FR_16: As a manager, I want to be able to view and make changes to restaurant operations so that I can make adjustments to the employees, staff, seating arrangements, etc, accordingly.

FR_17: As a manager, I want to be able to view restaurant revenue by day, week, month, quarter, and year so that I can keep up with how the restaurant is doing financially.

FR_18: As a manager, I want to be able to see the percentage of revenue obtained for each menu item and which ones are most/least popular so that I can make changes to the menu as needed.

FR_19: As a manager, I want to be able to view average turnaround and preparation time statistics so that I can make improvements in how food is prepared and how customers are served.

FR_20: As a manager, I want to be able to view periods of high/low traffic so that I can ensure the appropriate number of staff are available during those times.

NFR_1: Tables on the display should be color-coded to signify if a table is occupied, dirty, or empty.

NFR_2: Table status updates and waiter assignments should populate within two seconds of being input into the system.

NFR_3: The system should be able to control multiple updates from different staff members simultaneously with no interruptions.

NFR_4: Usability tests should show that 95% of new users can complete basic actions within one minute without external help.

NFR_5: It should take no more than four button presses to complete common actions, such as entering orders and updates.

NFR_6: A list of a waiter's assigned tables should be generated for them within two seconds of logging into the system.

NFR_7: A UI of the menu should be integrated into the system to make entering orders simple and quick to do.

NFR_8: An expected ETA of a customer order should be kept by the order in the system and be dynamically updated every five minutes or when a cook manually updates it.

NFR_9: The UI should allow for multiple options of payment, including cash, credit card, and debit card.

NFR_10: There should be a queue system available for the cooks that allows them to see customer orders from earliest to latest.

NFR_11: There should be a function to allow cooks to enter ETA's for each customer order that will be updated so that the assigned waiter can view it.

NFR_12: When an order is marked as complete, a pop up notification should be sent to the assigned waiter's account to alert them of the order within two seconds.

NFR_13: A log of all restaurant employees should be kept that is only available to the manager(s).

NFR_14: Employee information should consist of personal information and have the ability to be modified.

NFR_15: Employees should be able to be archived or new profiles can be added.

NFR_16: Employee schedules and clock in/clock out times updated daily should be viewable in their profiles.

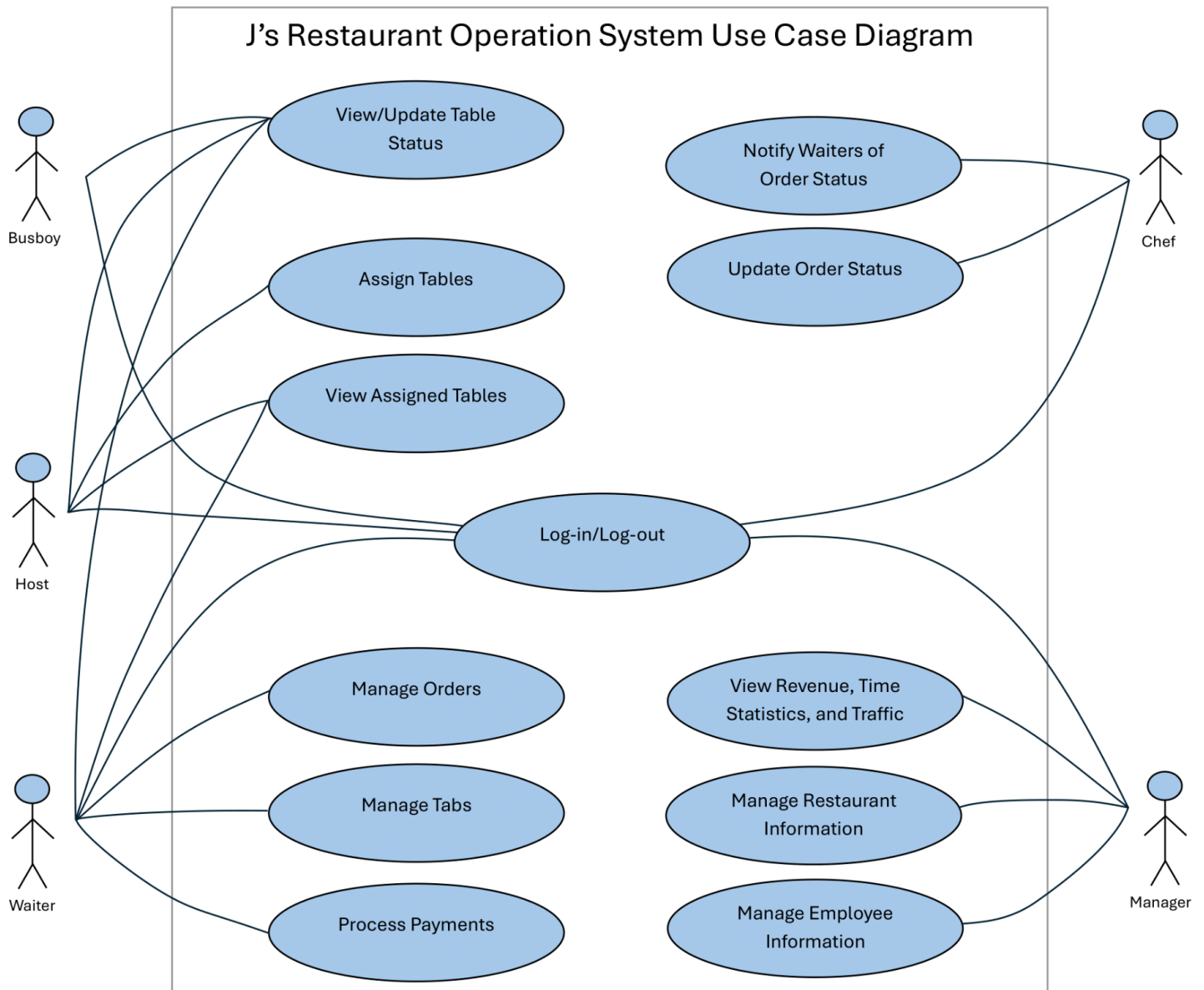
NFR_17: Menu changes and seating floor arrangements should be able to be modified by the manager to adjust to restaurant needs.

NFR_18: Restaurant revenue should be accessible to the manager and organized into different settings, such as by day, week, month, and menu item. This should be updated daily and in real time.

NFR_19: Operational data should be kept and stored for the manager to be review and be updated weekly.

Requirement Specification Documents

Use Case Diagram



Use Case Flow of Events

1.0 Flow of events for the Manager Use Case

1.1 Preconditions

No preconditions exist.

1.2 Main Flow

The main flow begins with the user logging into the system with their credentials which includes their username or employee ID and their password. The system must verify that the password is correct (E-1). The user must then select one of the following: View Revenue, Time Statistics, and Traffic, Manage Restaurant Information, or Manage Employee Information, Log Out.

If View Revenue, Time Statistics, and Traffic is selected, the S-1: Time Statistics subflow is performed

If Manage Restaurant Information is selected, the S-2: Restaurant/Item Information subflow is performed

If Manage Employee Information is selected, the S-3: subflow is performed

If Log Out is selected, the use case ends

1.3 Subflows

S-1: Time Statistics

The system displays the current time statistics of the restaurant and the user is given three options: View Revenue Information, View Item Information, and Home

If the user selects View Revenue Information, the system displays the current revenue information for the restaurant (E-2).

If the user selects View Item Information, the S-2: Restaurant Information subflow is performed.

If the user selects Home, the program takes the user back to the Main Flow.

S-2: Restaurant/Item Information

The system displays each item on the menu, its revenue, and its popularity. The user is also given two options: Add Menu Item (E-3) and Delete Menu Item (E-4).

If the Add Menu Item is selected, the program displays the different categories of the menu for the user to select. Once a category is selected, the user must type the name of the new item and select "Add Item."

If the Delete Menu Item is selected, the program displays the different categories of the menu for the user to select a specific item. Once an Item is selected, the user must press "Delete Item."

S-3: Employee Information

The program displays all of the employees at J's allowing the user to select each one to see all of their information. The user has three options: Add New Employee (E-5), Select Employee, and Home.

If the user selects Add Employee, the program will ask for new employee information.

If the user selects Select Employee, the program will display the existing employees information, schedule, and activity (E-6). The user has the option to delete selected employees from the system (E-7).

If the user selects Home, the user is taken back to the Main Flow.

1.4 Alternative Flows

E-1: Invalid Password

The system cannot find the given password linked to an employee in the system. The use case displays an error message and asks the user to try again.

E-2: Cannot Display Revenue

The system cannot access/display revenue information. The use case goes back to the home page.

E-3: Unable to Add Item

The system cannot add new items to the menu. The use case begins again.

E-4: Unable to Delete Item

The system cannot delete existing menu items. The use case begins again.

E-5: Unable to Add New Employee

The system cannot store the given employee information. The use case begins again.

E-6: Unable to Display Information

The system cannot access the selected employees information, schedule, or activity. The use case begins again.

E-7: Unable to Delete Existing Employee

The system cannot delete existing employee information. The use case begins again.

2.0 Flow of events for the Host Use Case

2.1 Preconditions

No preconditions exist.

2.2 Main Flow

The main flow begins with the user logging into the system with their credentials which includes their username or employee ID and their password. The system must verify that the password is correct (E-1). The program then displays the color coded floor plan so the user can see the assigned, free, and dirty tables. The user has the option to select tables/seats from the floor to update the table/seat status and assign wait staff.

If the user decides to update the tables status, the S-1: Update Table is performed

If the user selects Home, the use case will return to the floor plan.

If the user selects Log Out, the use case ends.

2.3 Subflows

S-1: Update Tables

The program gives the user options to update the table status: "Ready", "In Use", and "Dirty."

If the user selects "Ready", the user is prompted to assign a waiter to the table by entering their name or employee ID(E-2, E-3).

Once an option is selected, the program displays the updated floorplan.

2.4 Alternative Flows

E-1: Invalid Password

The system cannot find the given password linked to an employee in the system. The use case displays an error message and asks the user to try again.

E-2: Invalid Credentials

The system cannot find the given employee name or ID. The use case displays an error message and asks the user to try again.

E-3: Invalid Staff Option

The system cannot find the wait staff with a given employee name or ID. The use case displays an error message and asks the user to try again.

3.0 Flow of events for the Waiter Use Case

3.1 Preconditions

The Assign Tables subflow of the Host use case must be complete before this use case begins.

3.2 Main Flow

The main flow begins with the user logging into the system with their credentials which includes their username or employee ID and their password. The system must verify that the password is correct (E-1). The program then displays the floor plan. The program has a list of tables the user is assigned to along with three options for the user to select from: Manage Orders, Manage Tabs, and Process Payments.

If the user selects Manage Orders, the S-1: Orders is performed.

If the user selects Manage Tabs, the S-2: Tabs is performed.

If the user selects Process Payments, the S-3: Payments is performed.

If the user selects Home, the program returns to the Main Flow screen.

If the user selects Log Out, the use case ends.

3.3 Subflows

S-1: Orders

The program displays all the orders the user is assigned to with the option to create a new order. The user also has the option to select an existing order to check on its status or add more menu items.

If the user selects Create New Order, the program will display the menu for the user to select items from the menu. Once the order is complete (E-2), the user can send it to the kitchen (E-3).

S-2: Tabs

The program displays all the orders the user is assigned to. The author can select an order to finalize the tab and continue to the payment process.

S-3: Payments

The program asks the user for a payment method, cash or card. If the user selects card, the program will ask for the card information including the card number, expiration date(E-4), and security code. If the user selects cash, the program will prompt the user to insert the cash into the machine.

3.4 Alternative Flows

E-1: Invalid Password

The system cannot find the given password linked to an employee in the system. The use case displays an error message and asks the user to try again.

E-2: Unable to Add New Order

The system cannot create new order. The use case will begin at Order (S-1) subflow.

E-3: Unable to Send Order

The system cannot send orders. The use case will begin at Order (S-1) subflow.

E-4: Invalid Card

The given card information is incorrect/invalid. The use case will begin at the Payments (S-3) subflow.

4.0 Flow of events for the Chef Use Case

4.1 Preconditions

The Orders subflow of the Waiter use case must run before this use case can begin.

4.2 Main Flow

The main flow begins with the user logging into the system with their credentials which includes their username or employee ID and their password. The system must verify that the password is correct (E-1). The program then displays all of the orders placed in the order they were sent to the kitchen. Each order will have a field stating how long it has been since the order was placed and a field to update the order status.

If the user selects Update Order Status, the S-1: Update Status subflow is performed.

If the user selects Log Out, the use case ends.

4.3 Subflows

S-1: Update Status

The program asks the user to select an option to update the status of the order. The options include: In Progress and Ready.

If the user selects In Progress, the status of the order will be updated(E-2) to “In Progress.”

If the user selects Ready, the status of the order will be updated to “Ready” and the waiter who placed the order will be notified.

4.4 Alternative Flows

E-1: Invalid Password

The system cannot find the given password linked to an employee in the system. The use case displays an error message and asks the user to try again.

E-2: Unable to Update Status

The program cannot update order status. The use case will go back to the Main Flow.

5.0 Flow of events for the Busboy Use Case

5.1 Preconditions

No preconditions exist.

5.2 Main Flow

The main flow begins with the user logging into the system with their credentials which includes their username or employee ID and their password. The system must verify that the password is correct (E-1). The program then displays the color coded floor plan so the user can see the assigned, free, and dirty tables. The user has the option to select tables/seats from the floor to update the table/seat status.

If the user selects a seat/table, the S-1: Update Table Status (E-2) subflow is performed.

If the user selects Log Out, the use case will end.

5.3 Subflows

S-1: Update Table Status

The user has the option to update table status to “Ready”, “In Use”, or “Dirty”. When an option is selected that table’s/seat’s status will be updated accordingly.

5.4 Alternative Flows

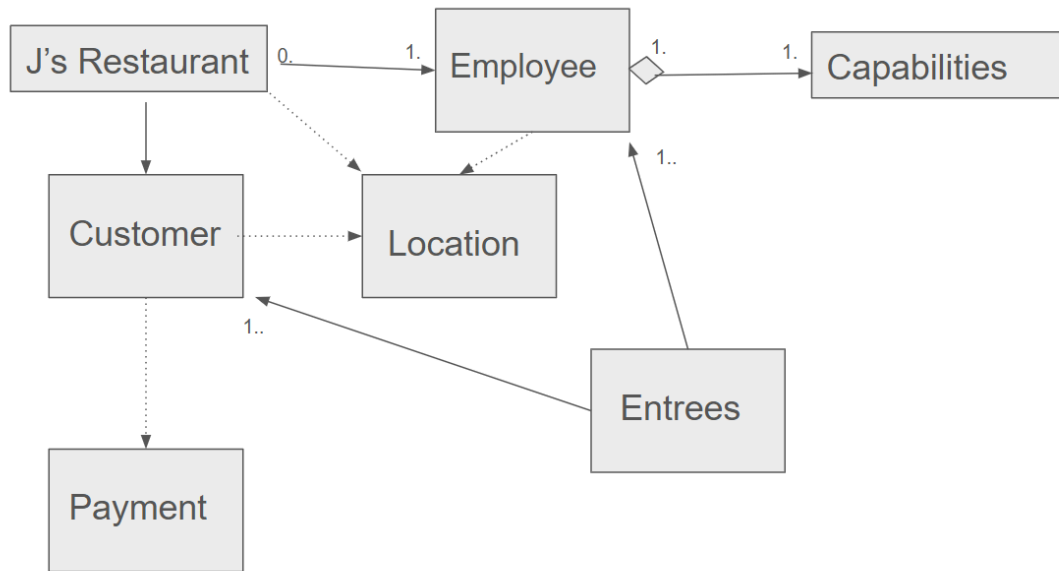
E-1: Invalid Password

The system cannot find the given password linked to an employee in the system. The use case displays an error message and asks the user to try again.

E-2: Unable to Update Table Status

The program is unable to update the selected table’s status. The use case will return to Main Flow.

Class Diagrams



Class Documentation

J's Restaurant class is one of the main classes that starts the entire program. It grants access to the user if the user entered the correct credentials. After logging in, the class separates into Employee or Customer depending on its setting. The class can also extract information from both customer and Employee classes if required.

The Employee class contains information for all of its employees. The employees' skill sets and daily accomplishments are sent to the capabilities class where that information is also extracted to update employees' information. The class also stores the employees' address using the location class. It also receives information from the entrees class to keep the employee up to date.

The Capability class only works by receiving updates from the Employee class. It stores the employees' skill sets and their accomplishments. The Class also sends that information back to the employee class to update any data in the Employee class.

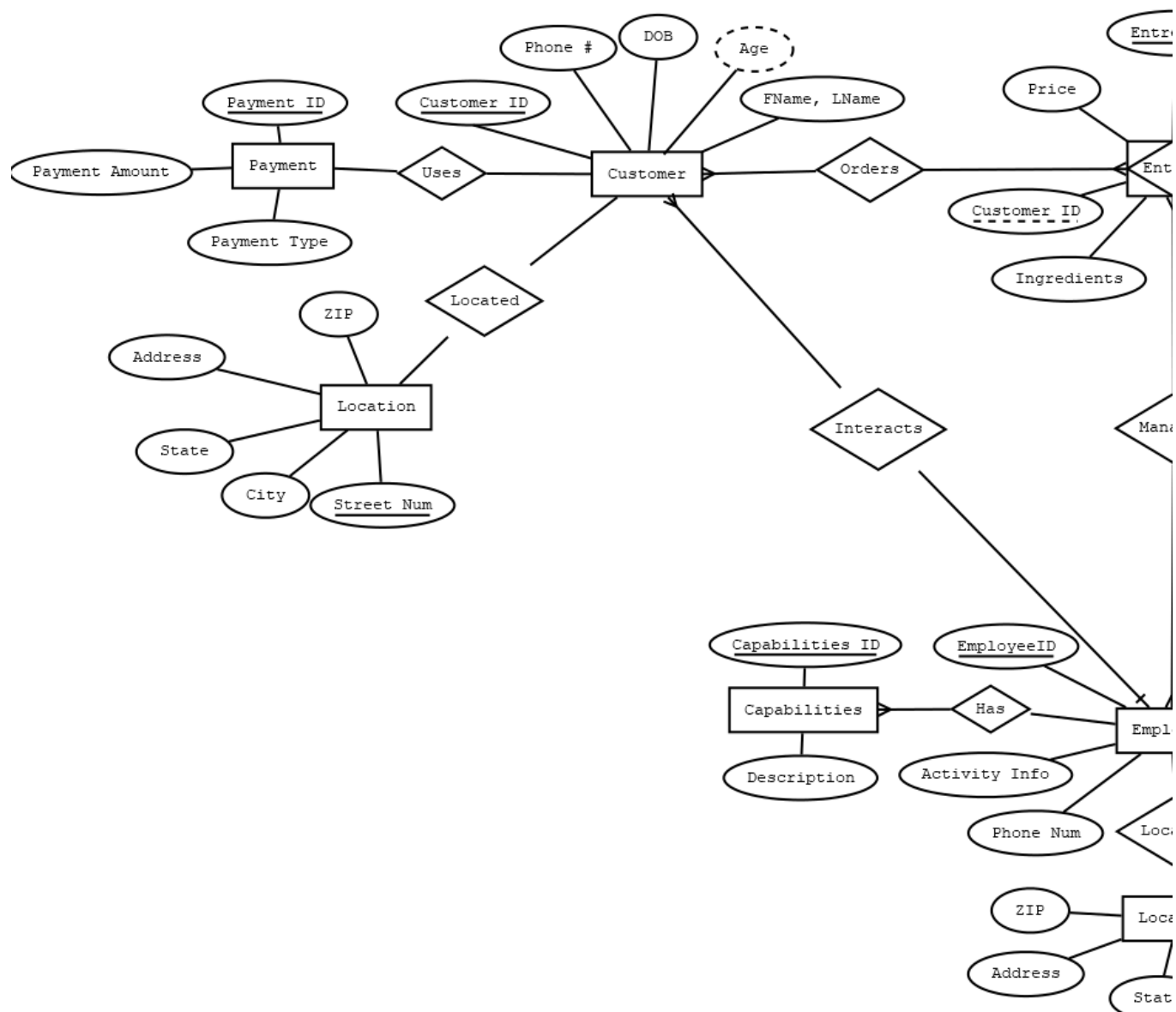
The Customer class starts by the customer checking into the restaurant. After checking in, the customer also enters their address which is stored through the location class. The Class receives the menus for the customer from the entree class and sends the chosen menu back to the entree class. Once the customer finishes the order, the entree class sends the cost for the entree back to the customer class. The customer class then sends that information to the payment class.

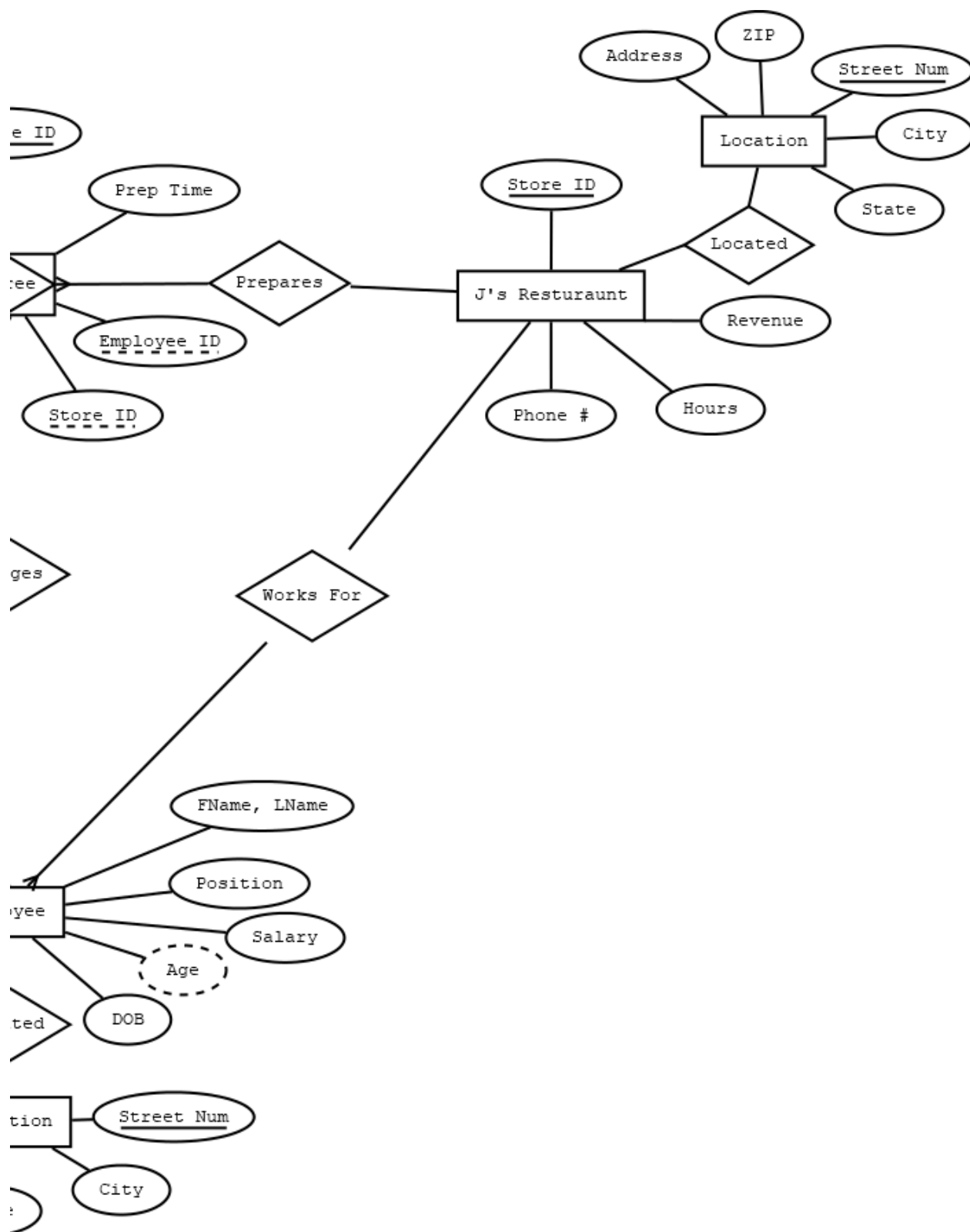
The payment class receives information from the customer class and shows it to the user. Once the user pays, that information is stored until called upon by other classes.

The location class helps store the address for both employees and customers. It receives the information then stores it based on whether the information is from an employee or a customer. It can also send the information back if called upon.

The entree class contains information on all the menus of the restaurant. It constantly sends the menu to both customer and employee classes and updates based on those two classes.

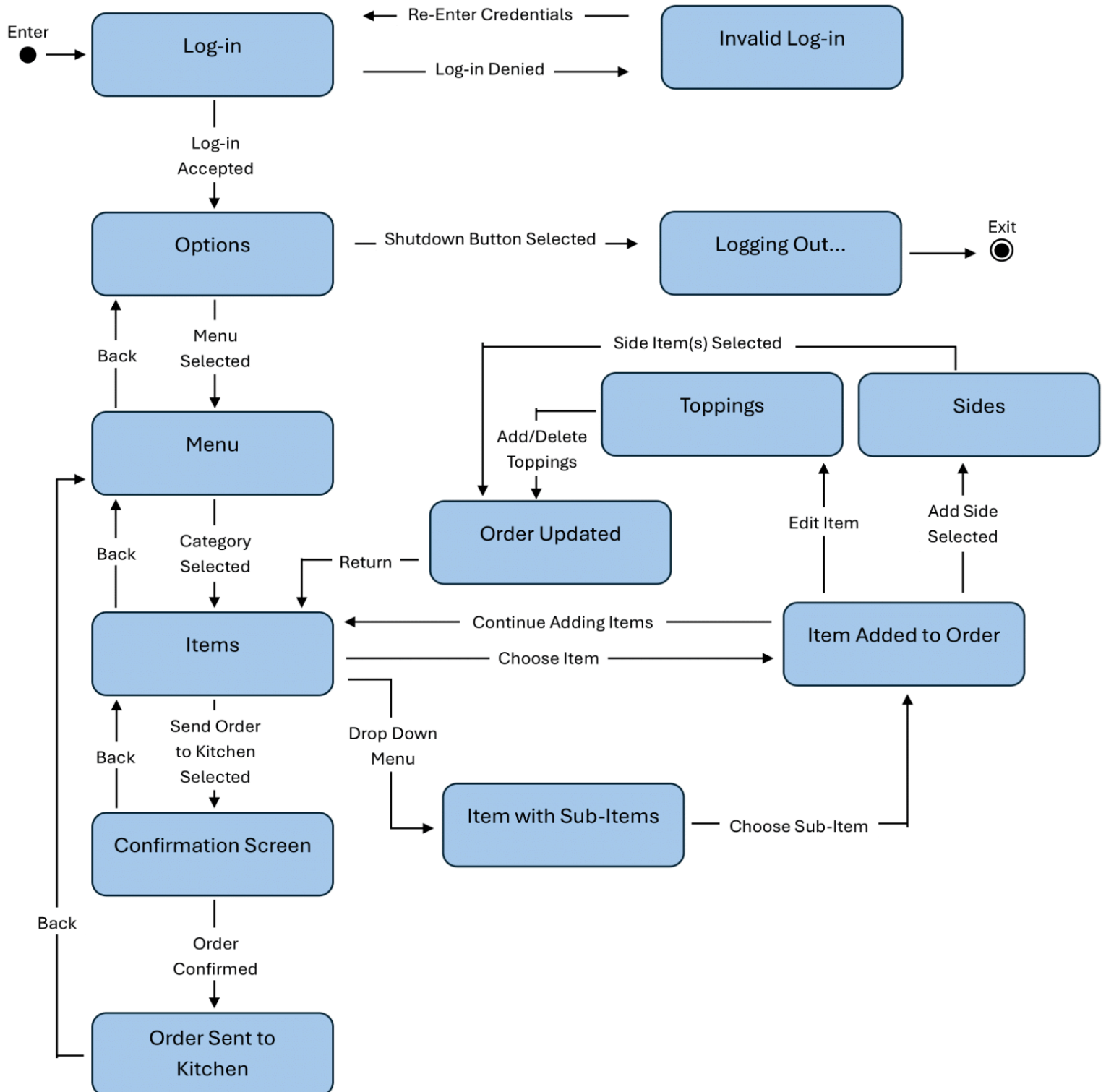
ER Diagram



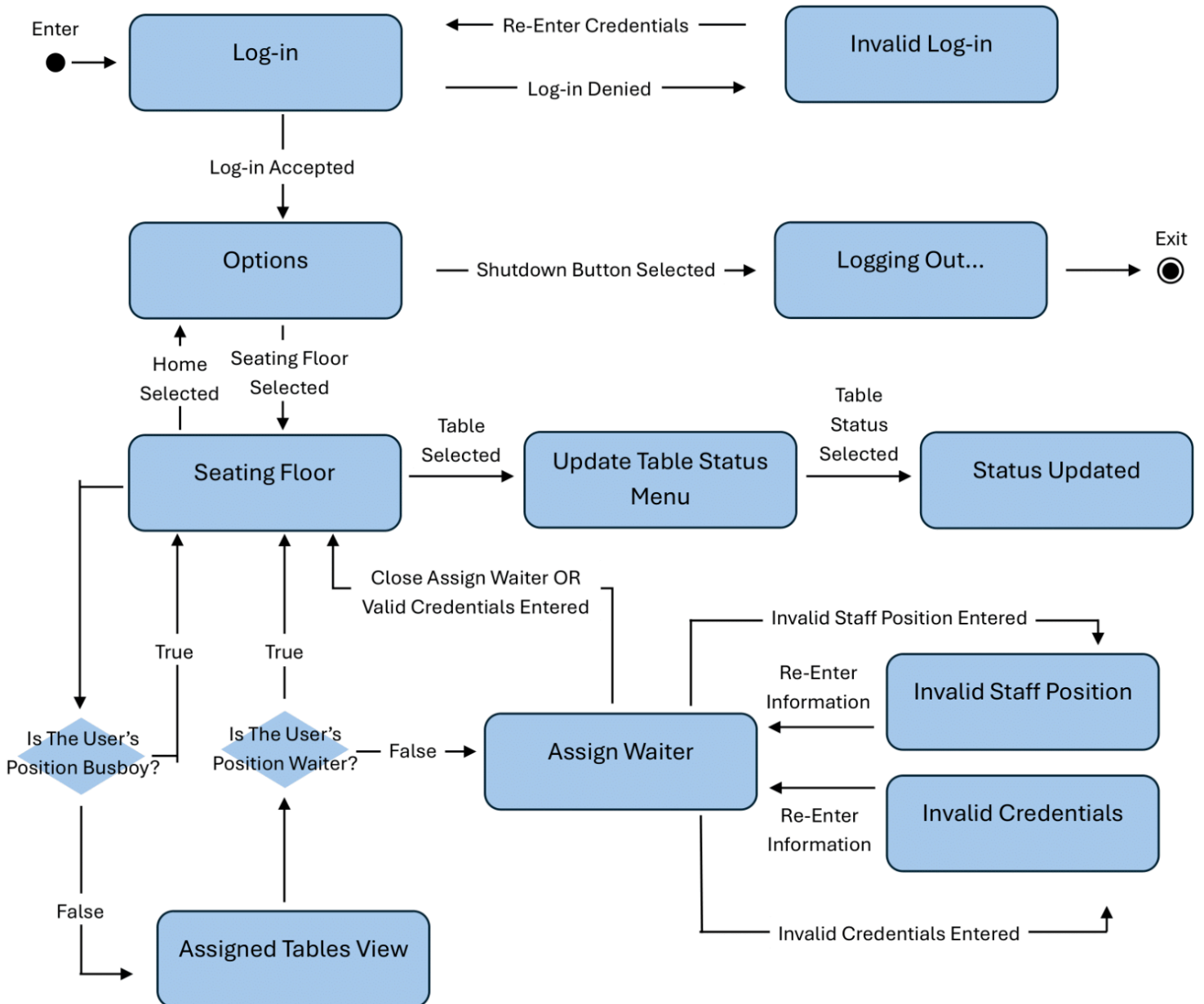


State Transition Diagram

Creating Orders



Managing Tables



Managing Employee Information

